

Arbitrum Builder Pods

Assignment

 Deadline Today	 Submit Via GitHub	 Build With AI Assistance
--	---	--

Objective

Strengthen your foundational understanding of Web3 concepts by building a multi-page website from scratch. This bridges the gap between theory and hands-on implementation before you move deeper into Web3 development.

Note: Build the entire website with the help of AI (any AI-native IDE you prefer).

Reference PPT: <https://canva.link/mgiy5on2c9bzljf>

Task : Build a 4-Page Website

Your entire assignment is a single deliverable: one cohesive website that contains four pages. Each page covers a distinct Web3 concept as a functional mini-app or informational module.

The four pages to build:

Page 1 : Home / Landing

Responsive landing page introducing your chosen Web3 theme with hero, features, and footer sections.

Page 2 : Concepts

Visual comparison cards covering core Web3 concepts (Web2 vs Web3, Ethereum vs Bitcoin, etc.)

Page 3 : Live Prices

Live crypto price dashboard fetching real-time ETH and BTC data from the CoinGecko API.

Page 4 : Block Simulator

Interactive block mining simulator that demonstrates hashing, nonces, and chain immutability.

Why one website?

Combining all four pieces into a single navigable site teaches you real project structure - how pages link together, how a codebase is organized, and how to ship something end-to-end. This is how real products are built.

Page 1 - Home / Landing

Build the main landing page of the website. Pick exactly one theme and make the entire page feel cohesive - not like multiple sections stitched together.

Pick One Theme

- Blockchain / Web3 Introduction
- Visual representation of how blocks are connected in a blockchain
- Explainer for transactions, blocks, and chains
- Arbitrum or Layer 2 overview

Required Sections

Header	Site name / logo and navigation links to all four pages
Hero / About	Clearly introduce the topic with a headline and short description
Features / Benefits	At least 3 key points with icons or visuals
Footer	Your name, GitHub link, batch name

If You Pick the “Arbitrum or Layer 2 overview” Theme

Include a short section on this page explaining:

- Why Ethereum needed Layer 2 solutions
- What Arbitrum is and how it helps
- One real-world benefit of using Arbitrum over mainnet

**Tip**

If you pick the Arbitrum theme, you already have everything you need from the sessions. Use that knowledge directly here.

Page 2 - Concepts

Build a styled page that presents core Web3 concepts as visual comparison cards or tables. This is your reference page - the first thing visitors see that explains the fundamentals.

Concepts to Cover

- Web2 vs Web3
- Ethereum vs Bitcoin
- Public Key vs Private Key
- Blockchain vs Traditional Databases

Layout Requirements

- Use a card, table, or side-by-side layout for each concept - no plain paragraphs
- Write explanations in your own words - simple, clear, and concise
- The page should feel like an educational explainer, not a textbook dump

Why does this matter ?

Writing concepts into a visual layout forces deeper understanding. Anyone can paste a definition - building a comparison card means you truly understood both sides.

Page 3 - Live Prices

Build a live cryptocurrency price dashboard on this page. Fetch real-time data from the CoinGecko public API - no API key required.

Required

- Fetch live prices for at least ETH and BTC from CoinGecko
- Display them in styled cards showing: coin name, current price (USD), 24h price change %
- Show a green up arrow or red down arrow based on whether price change is positive or negative
- Add a Refresh button to re-fetch data on demand

Suggested API Endpoint

```
https://api.coingecko.com/api/v3/simple/price?ids=ethereum,bitcoin&vs_currencies=usd&include_24hr_change=true
```

Optional Enhancements

- Add more coins (e.g., SOL, MATIC, ARB)
- Show a mini sparkline or price trend indicator
- Add a search bar to look up any coin (optional)

Why CoinGecko?

No API key needed, completely free, and directly relevant to Web3. Fetching ETH/BTC prices is a rite of passage for every Web3 developer.

Page 4 - Block Simulator

Build an interactive block mining simulator on this page using plain JavaScript. This is the most hands-on way to understand nonce, hashing, and chain integrity - the concepts from the sessions made tangible.

What to Build

- Input fields for: Block Data (text), Previous Hash (hardcoded or auto-filled), Nonce (number)
- A Mine button that increments the nonce until the hash output starts with '00' (simulated proof-of-work)
- Display the resulting hash and show Block Valid or Block Invalid
- A second block that reads the hash from Block 1 as its Previous Hash - showing the chain
- If you change Block 1's data, Block 2 should automatically turn invalid

Important Notes

- No blockchain library needed - pure JavaScript is sufficient
- Use a simple string-based fake hash or the built-in Web Crypto API (SHA-256)
- The chain-breaking effect when Block 1 changes is the key insight of this page

The Core Insight

When you change data in Block 1, Block 2 breaks. This is immutability in action. Building this with JavaScript will teach you more than any explanation ever could.

Navigation & Structure

All four pages must be connected via a shared navigation bar. A visitor should be able to move between pages without going back to the repo or manually editing the URL.

Minimum Navigation Requirements

- A header or nav bar present on every page
- Links to all four pages accessible from every page
- Active page visually highlighted in the nav
- Consistent styling across all pages (same fonts, color palette, spacing)

Tech Stack

- HTML / CSS / JavaScript - perfectly acceptable for all four pages
- React or Next.js - great choice if you want routing and component reuse
- Either way, all four pages should live in one repository

Submission Guidelines

Submit all of the following:

1. GitHub repository containing the complete website (all four pages)
2. Screenshots - at least one screenshot per page (four total)
3. A README.md file in the root of the repository

README.md Should Include

- Project name and a short description of the website
- List of all four pages and what each one does
- How to install and run the project locally
- Any known issues or things you would improve

Evaluation Criteria

Criterion	What We Look For
Clarity of Concepts (Page 1)	Are Web3 ideas explained accurately in your own words?
Functionality	Do all four pages run without errors?
Navigation & Cohesion	Do all pages feel like one unified website, not four separate files?
UI Structure & Responsiveness	Is the layout clean, organized, and mobile-friendly?
Code Organization	Is your code readable, commented where needed, well-structured?
Blockchain Simulation (Page 4)	Does the simulator correctly reflect real Web3 logic?
Overall Effort	Does the submission show genuine engagement with the material?

Timeline

Event	Date
Submission Deadline	Within 2 days

All the best!

Every line of code you write this week is a step toward building on Arbitrum.